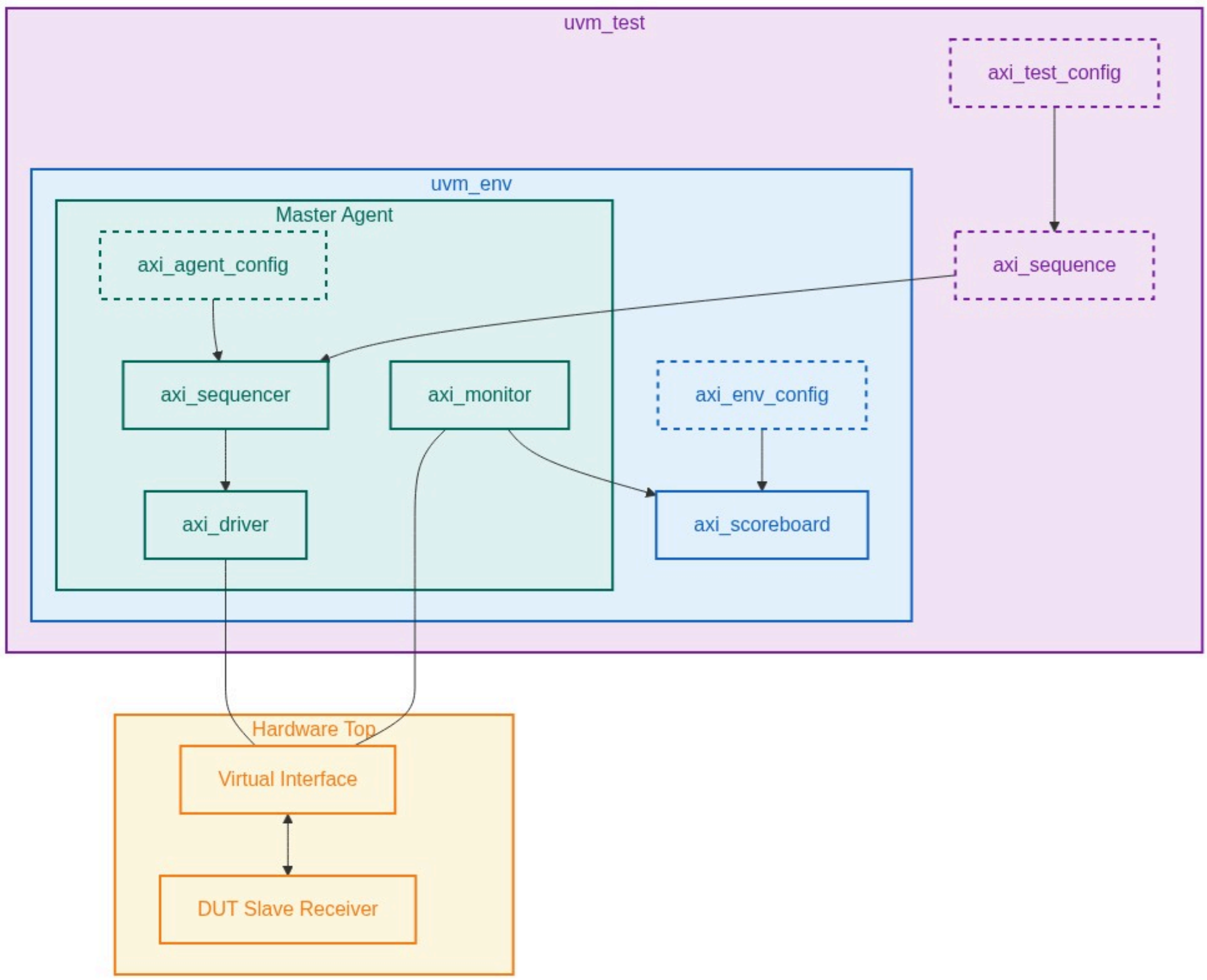
# Verification Plan Report

## Project: AXI-Stream Master VIP Verification Plan

Version: 9.0

Verification Role: Master

## 1. UVM Environment Architecture




## 2. Protocol Configurations

| Feature | Configuration Name | Impact |
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|---|---|---|
| Handshake Protocol | TVALID_TREADY_Async_Handshake | Functionality impact: Forces pipeline stalling and dictates stability bounds for the entire payload vector.<br>Performance impact: dictates maximum theoretical throughput in scenarios involving combinatorial TREADY backpressure. |
| Packet Framing | TLAST_Packet_Framing_Support | Functionality impact: Requires end-to-end exact reconstruction of packet boundary assertions to govern data routing layer and destination buffers.<br>Performance impact: dictates fragmentation penalties and dynamic arbitration thresholds. |
| Byte-Level Stream Semantics | TKEEP_TSTRB_Qualifier_Encoding | Functionality impact: necessitates robust reconstruction algorithms in the scoreboard to extract physical data while discarding positional or null bytes.<br>Performance impact: alters actual byte-consumption rate compared to clock-cycle metrics. |
| Reset Sequencing | ARESETn_Mid_Transfer_Isolation | Functionality impact: compels the testing environment to trigger state-machine destruction mid-transfer.<br>Verification constraint: requires the VIP driver to safely discard tracking state and the monitor to correctly sequence clean IDLE resets asynchronously. |
| Wakeup Signaling (AXI5) | TWAKEUP_Pre_Transfer_Signaling | Functionality impact: introduces an external deadlock dependency if TWAKEUP assertion lags TVALID assertion.<br>Performance impact: latency models must explicitly adjust for TWAKEUP lead times before TREADY is physically accessible. |
| Stream Identification | TID_TDEST_Interleaving_Routing | Functionality impact: forces the scoreboard multiplexer logic to concurrently maintain dynamically allocated data streams to track ordering compliance.<br>Performance impact: tests the DUT buffer saturation across varying interleave depths. |
| Parity Protection (AXI5) | Odd_Parity_Check_Generation | Functionality impact: requires single-bit error injection capabilities within the VIP transmission chain to exercise the DUT standard error logic.<br>Verification constraint: the monitor must passively track hardware parity matrices parallel to the native data payload across all TKEEP permutations. |

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| Continuous Packets Mode | Continuous_Packet_Constraint_Mode | Functionality impact: strictly constrains the randomization matrices of the VIP to avoid null-byte and inter-packet gaps. Performance impact: maximizes arbitration efficiency and validates peak continuous payload saturation without recovery cycles. |

## 3. Test Requirements

| Index | Name | Description |
|---|---|---|
| REQ_MST_01 | TVALID-to-Payload Stability Dependency | Direct signal dependency: When TVALID=1 and TREADY=0, the state of TDATA, TSTRB, TKEEP, TLAST, TID, TDEST,<br>and TUSER must not change. The protocol mandates that a Transmitter must unconditionally hold payload signals<br>stable until TREADY=1 is sampled coincidently with TVALID=1 to prevent race conditions during sampling.<br><br>Verification Intent: Ensure the VIP sequences drive stable payload signatures across highly randomized TREADY stall windows.<br>The protocol monitor must fire fatal assertions if any payload bit changes mid-stall. This enforces strict spec compliance<br>at the conformance boundary. |
| REQ_MST_02 | TKEEP to TSTRB Mask Constraint | Signal Relationship: TKEEP and TSTRB share a strict dependency. TKEEP=0 designates a null byte, making TSTRB mathematically<br>irrelevant. However, if TKEEP=0, asserting TSTRB=1 is an architecturally reserved condition. TKEEP=1/TSTRB=1 is a data byte;<br>TKEEP=1/TSTRB=0 is a position byte.<br><br>Verification Intent: Over-generate sparse transfer layouts. The VIP stimulus will enforce valid TKEEP/TSTRB combinations by<br>default. We will inject the illegal TKEEP=0/TSTRB=1 combination under a negative-stimulus configuration to verify<br>the standard DUT responds properly to corruption via its native error interfaces. |

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| REQ_MST_03 | TLAST to TVALID Synchronization | Dependency Model: TLAST acts as the packet termination qualifier and holds logical meaning ONLY when TVALID=1. TLAST assertion<br>cannot occur independent of TVALID. A packet starts immediately after a previous TLAST assertion (with TVALID=1), or on the<br>first TVALID=1 post-reset for a given TID/TDEST tuple.<br><br>Verification Intent: The VIP sequences will manipulate the interval between TVALID events and TLAST events, creating randomized<br>multi-beat packet formations up to 4096 beats, single-beat packets, and zero-data null packets. |
| REQ_MST_04 | TWAKEUP Pre-Transfer to TVALID/TREADY Lead Time | Signal dependency: TWAKEUP is an AXI5 preemptive signal. If TWAKEUP transitions HIGH concurrently with TVALID,<br>the spec dictates that TWAKEUP MUST remain HIGH until TREADY=1 completes the handshake. If TWAKEUP leads TVALID,<br>it provides power-activation buffering logic to the Slave.<br><br>Verification Intent: VIP constraints will sweep the temporal relationship: TWAKEUP asserted randomized cycles (1..32) prior<br>to TVALID, and TWAKEUP simultaneous with TVALID. The DUT must correctly un-stall its TREADY pathway in standard operating conditions. |
| REQ_MST_05 | TDATACHK to Null TKEEP Parity Computation | Intersection analysis: TDATACHK represents the odd parity vector of the TDATA payload. The spec strictly mandates that<br>the parity check MUST be driven across ALL physical byte lanes regardless of the TKEEP mask state. A TKEEP=0 lane<br>must still maintain active TDATA parity protection.<br><br>Verification Intent: The VIP transmitter passively calculates parity bits on all generated physical bus vectors independently<br>of the TKEEP/TSTRB configuration. Tests will randomize data underlying disabled TKEEP bytes to verify the DUT parity error check logic. |

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| REQ_MST_06 | ARESETn Active Suppression of TVALID | Global Override: ARESETn=LOW overrides all normal operations. TVALID must remain LOW while ARESETn is LOW. TVALID must only transition HIGH on a rising ACLK edge occurring strictly after the ACLK edge where ARESETn transitions HIGH.<br><br>Verification Intent: VIP forces asynchronous reset generation at peak bus congestion (TVALID=1, TREADY=0). The VIP automatically clamps TVALID=0. Monitors verify zero combinatorial bleed-through on TVALID after reset injection. |
| REQ_MST_07 | Arbitration Independence of TID Stream Paths | Stream interdependency check: TDATA payload transfers sharing matching TID/TDEST identifiers must maintain strict temporal ordering. Transfers with distinct TID identifiers hold zero temporal interdependency and may be interleaved at transfer granularity.<br><br>Verification Intent: The VIP runs a heavily randomized sequencer interleaving four unique TID streams across a shared physical driver. The scoreboard passively demultiplexes TIDs and checks order-invariant progression within each unique identifier. |

## 4. Test Cases & Mapping

| Index | Scenario | TR Mapping |
|---|---|---|
| TC_MST_001 | Directed Handshake Stalling: The VIP generator issues a directed 256-beat incremental stream. Randomization logic dynamically injects 0-to-30 cycle stalls on VIP TVALID drops between beats. The standard DUT is observed for robust pipeline progression under starvation. | REQ_MST_01 |
| TC_MST_002 | Directed Combinatorial Backpressure: VIP drives back-to-back continuous payloads. Randomization is applied to the DUT's simulated TREADY latency to generate 0-wait to 10-wait boundary backpressures, verifying strict VIP TVALID adherence. | REQ_MST_01 |
| TC_MST_003 | Directed Sparse Byte Array: VIP generates a directed 4096-beat continuous sequence where TKEEP is randomized per byte lane. Constraint guarantees legal TSTRB propagation. Scoreboard tracks physical byte offset mapping. | REQ_MST_02 |

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| TC_MST_004 | Negative TSTRB Qualifier Corruption: VIP is directed to force an illegal TKEEP=0, TSTRB=1 array mask on byte lane 3. VIP fault injection completes safely. DUT recovery and fault reporting logic evaluates the standard protocol violation natively. | REQ_MST_02 |
| TC_MST_005 | Directed Null Packet Generation: VIP randomly configures packet sizes, selecting Null-Size (TKEEP=all zeros, TLAST=1, TVALID=1) interspersed with 5-beat valid data payloads. Observes DUT buffer extraction handling. | REQ_MST_03 |
| TC_MST_006 | Directed Wakeup Latency Sweep: VIP sequences TWAKEUP generation ahead of TVALID randomly partitioned across [0-to-15] ACLK cycles. Simulates dynamic power-lane synchronization against the DUT logic. | REQ_MST_04 |
| TC_MST_007 | Directed Checksum Null-Lane Sweep: VIP directed to transmit 1000 transfers where TKEEP=0 over 50% of the byte lanes. VIP calculates valid parity. Monitor observes parity conformance across inactive physical dimensions. | REQ_MST_05 |
| TC_MST_008 | Negative Parity Injection: VIP explicitly flips the checksum parity bit on TDATACHK[2] dynamically during a valid active transmission. DUT Parity reporting interrupt is interrogated without custom override commands. | REQ_MST_05 |
| TC_MST_009 | Directed Reset Destruction: VIP initiates concurrent multi-stream traffic. On randomized ACLK edge, VIP forcefully asserts ARESETn. Verifies TVALID collapse latency and clean post-reset VIP spin-up against standard DUT. | REQ_MST_06 |
| TC_MST_010 | Directed TID Fragmentation: VIP generates four independent TID streams. The driver sequencing logic is randomized to fragment transfers at the single-beat level, alternating TID IDs constantly. Scoreboard confirms stream fidelity. | REQ_MST_07 |

## 5. Functional Coverage Groups

| **Group Name** | **Description** |
|---|---|
| cg_tvalid_tready_temporal_cross | Multi-dimensional coverage cross-referencing TVALID lead times with TREADY hold times across 5 distinct duration bins. |
| cg_tkeep_tstrb_combinatorial | Coverage metric tracking standard bitwise legal interactions against negative fault injections. |
| cg_tlast_tvalid_cadence | Bins corresponding to TLAST positional depth against packet volumes, tracking zero-data packets dynamically. |
| cg_twakeup_lead_bins | Array of tracked intervals validating coverage across all possible pre-transfer timing limits specified in testing. |

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| cg_tdatachk_masked_integrity | Parity sequence coverage cross-referencing TDATACHK logic states mapped explicitly against TKEEP=0 vector elements. |
| cg_aresetn_asynchronous_window | State-machine phase bins where the reset logic was forcefully injected by the VIP. |
| cg_tid_fragmentation_depth | Crosses interleaved Stream IDs with maximum consecutive beats, confirming extreme edge permutation boundary interleaving. |

## 6. Interface Signals

| Signal Name | Direction | Width | Description |
|---|---|---|---|
| ACLK | Input (to VIP Master and DUT Slave) | 1 | Global synchronization clock. |
| ARESETn | Input (to VIP Master and DUT Slave) | 1 | Active-LOW global reset. |
| TVALID | Output (VIP Master / Transmitter) | 1 | Driven by the VIP Master. Indicates a valid transfer payload is present. |
| TREADY | Input (from DUT Slave / Receiver) | 1 | Driven by the DUT Slave to signal readiness to accept a transfer. |
| TDATA | Output (VIP Master / Transmitter) | TDATA_WIDTH | Primary data payload. |
| TSTRB | Output (VIP Master / Transmitter) | TDATA_WIDTH/8 | Per-byte data qualifier. |
| TKEEP | Output (VIP Master / Transmitter) | TDATA_WIDTH/8 | Per-byte position qualifier. |
| TLAST | Output (VIP Master / Transmitter) | 1 | Packet-boundary indicator. |

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| TID | Output (VIP Master / Transmitter) | TID_WIDTH | Stream identifier. |
| TDEST | Output (VIP Master / Transmitter) | TDEST_WIDTH | Routing identifier. |
| TVALIDCHK | Output (VIP Master / Transmitter, AXI5-Stream only) | 1 | Odd parity check bit for TVALID. |
| TDATACHK | Output (VIP Master / Transmitter, AXI5-Stream only) | TDATA_WIDTH /8 | Odd parity check per byte lane of TDATA. |
| TLASTCHK | Output (VIP Master / Transmitter, AXI5-Stream only) | 1 | Odd parity check bit for TLAST. |
| TREADYCHK | Input (from DUT Slave / Receiver, AXI5-Stream only) | 1 | Odd parity check bit for TREADY, generated by the DUT Slave. |

## 7. Verification Environment Structure

### Root: axi_stream_master_vip_tb/

|-- master_agent : VIP Master Agent: Sequencer, Driver (drives TVALID/TDATA/TLAST/parity), Monitor
|-- env : UVM Environment: Scoreboard, Protocol Checker, Coverage Collector, EnvConfig
|-- sequences : Stimulus library: handshake variants, sparse-stream, back-to-back, null-packet, parity-gen, parity-inject, reset, i
|-- tests : UVM test classes mapping each scenario to sequence combinations and config overrides
|-- assertions : SVA property file for real-time protocol conformance (stability, reserved-encoding, parity)
|-- top : Simulation top: DUT (Slave) instantiation, virtual interface binding, clock and reset generation
|-- verif_plan : Artifacts: YAML traceability matrix, PDF report, sub-prompt reference files